

Revision del dataset banknote paso a paso

Mismo esquema que con Iris, una tabla por paso en el workspace

```
clear; close all; clc
```

Paso 0: dato crudo

```
datos = load('data_banknote_authentication.txt');
T = array2table(datos(:,1:4), 'VariableNames',
{'varianza', 'asimetria', 'curtosis', 'entropia'});
T.clase = categorical(datos(:,5), [0 1], {'autentico', 'falso'});
vars = T.Properties.VariableNames(1:4);
T0_crudo = T;
summary(T0_crudo)
```

T0_crudo: 1372x5 table

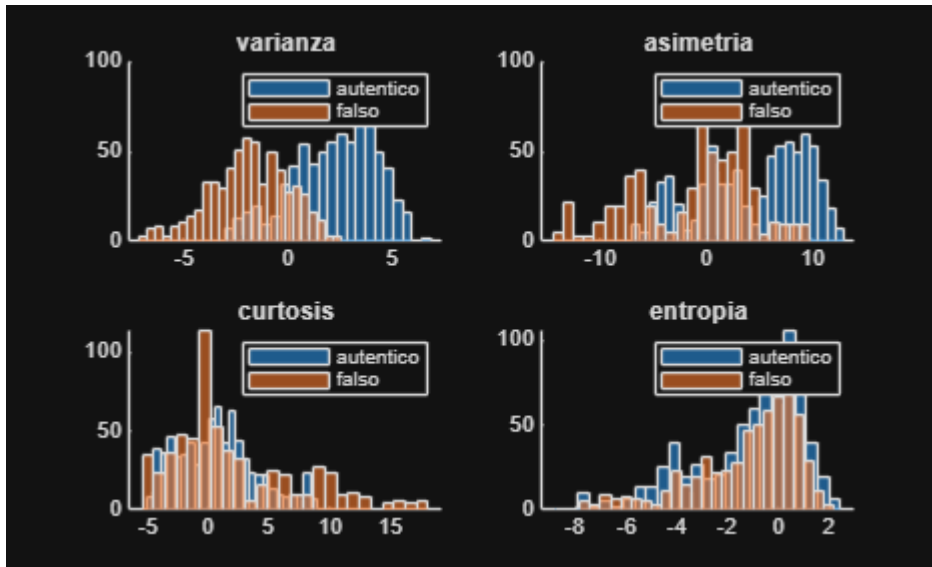
Variables:

```
varianza: double
asimetria: double
curtosis: double
entropia: double
clase: categorical (2 categories)
```

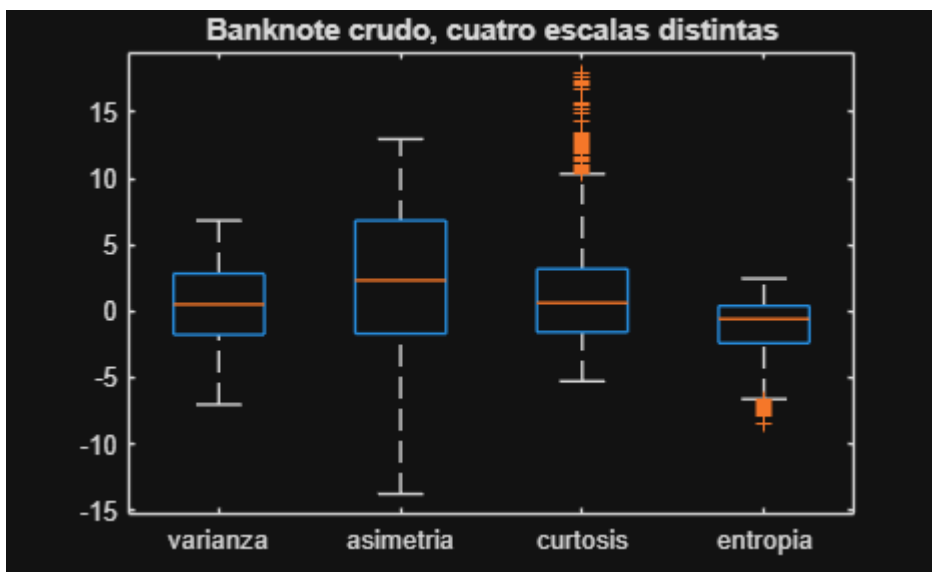
Statistics for applicable variables:

	NumMissing	Min	Median	Max	Mean	Std
varianza	0	-7.0421	0.4962	6.8248	0.4337	2.8428
asimetria	0	-13.7731	2.3197	12.9516	1.9224	5.8690
curtosis	0	-5.2861	0.6166	17.9274	1.3976	4.3100
entropia	0	-8.5482	-0.5867	2.4495	-1.1917	2.1010
clase	0					

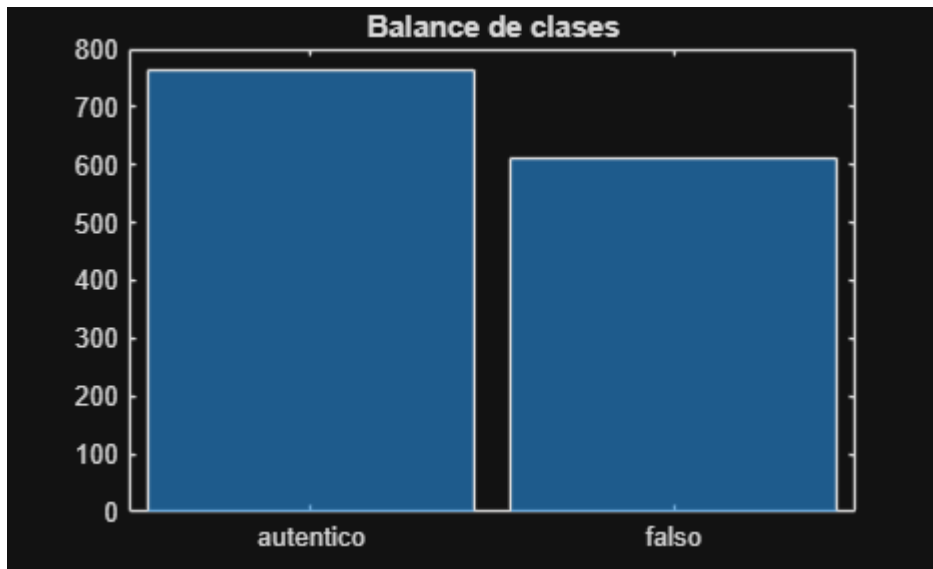
```
figure
for k = 1:4
    subplot(2,2,k); hold on
    for c = categories(T.clase)'
        histogram(T.(vars{k})(T.clase == c{1}), 25, 'DisplayName', c{1})
    end
    title(vars{k}); legend
end
```



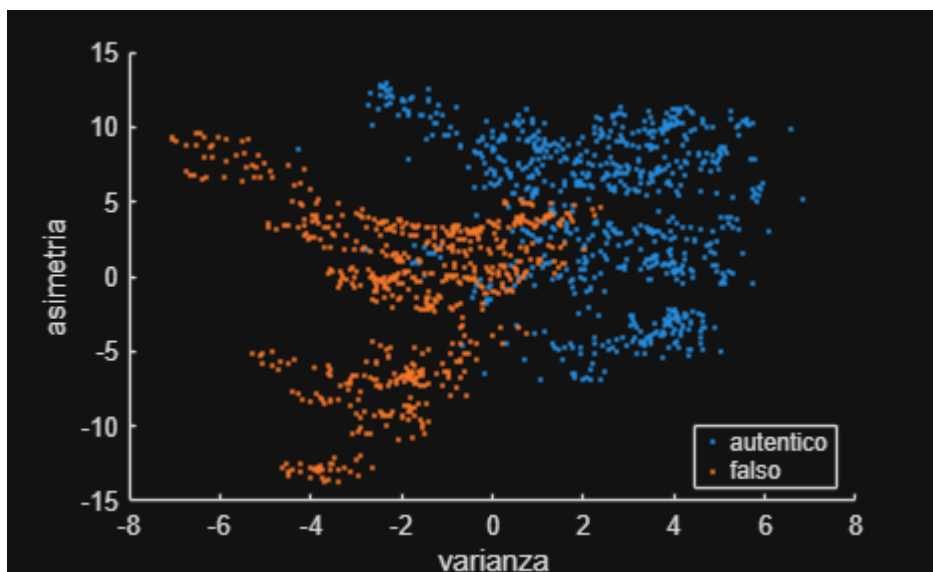
```
figure
boxplot(T[:,vars], 'Labels', vars); title('Banknote crudo, cuatro escalas
distintas')
```



```
figure
histogram(T.clase); title('Balance de clases')
```



```
figure
gscatter(T.varianza, T.asimetria, T.clase)
xlabel('varianza'); ylabel('asimetria')
```



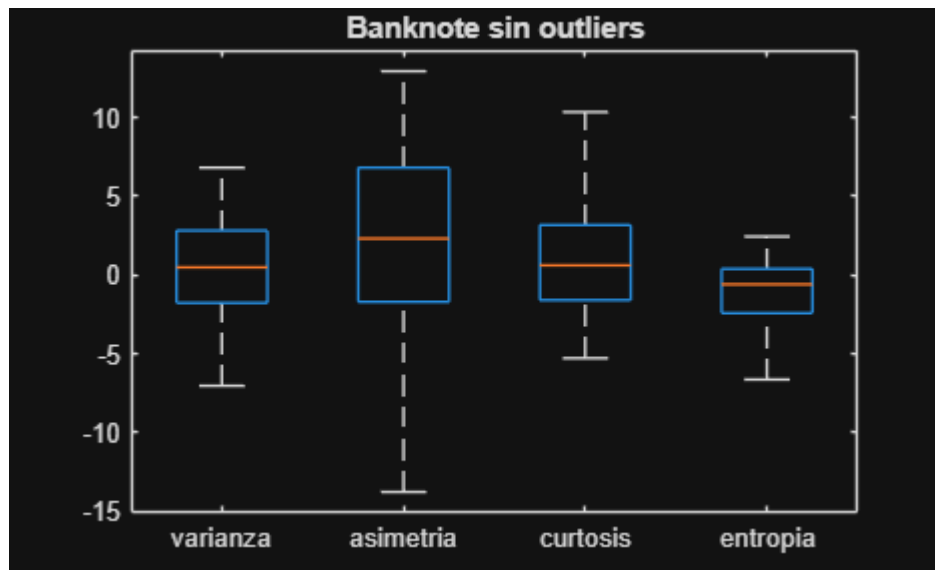
Paso 1: outliers con la regla 1.5*IQR, recortando al borde

```
T1_outliers = T0_crudo;
for k = 1:4
    x = T1_outliers.(vars{k});
    q = quantile(x, [0.25 0.75]);
    li = q(1) - 1.5*(q(2)-q(1)); ls = q(2) + 1.5*(q(2)-q(1));
    fuera(k) = sum(x < li | x > ls);
    x(x < li) = li; x(x > ls) = ls;
    T1_outliers.(vars{k}) = x;
end
```

```
fuera
```

```
fuera = 1x4
      0      0      57      33
```

```
figure
boxplot(T1_outliers{:,vars}, 'Labels', vars); title('Banknote sin outliers')
```



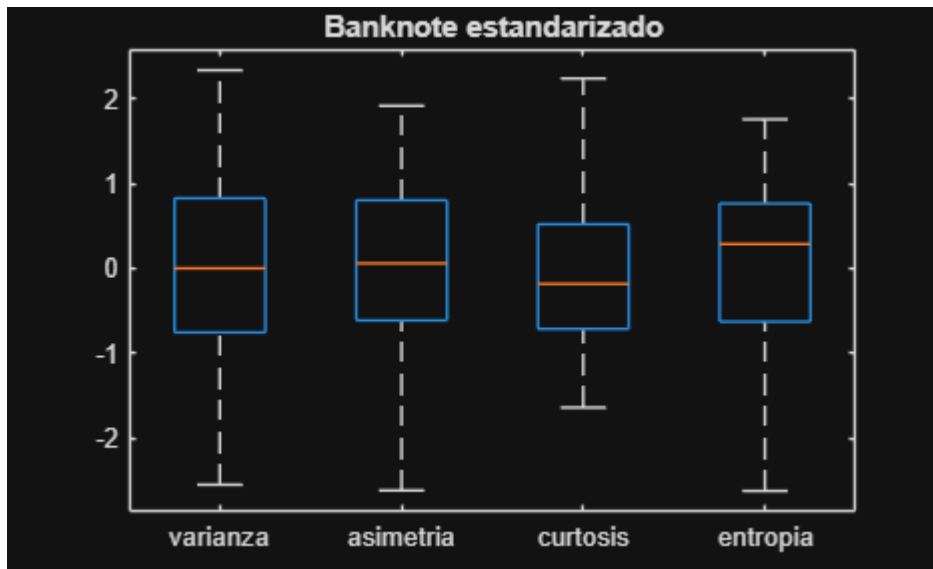
Paso 2: balanceo, 610 autenticos al azar para igualar a los 610 falsos

```
rng(1)
falsos = T1_outliers(T1_outliers.clase == 'falso', :);
autenticos = T1_outliers(T1_outliers.clase == 'autentico', :);
autenticos = autenticos(randperm(height(autenticos), height(falsos)), :);
T2_balanceado = [autenticos; falsos];
countcats(T2_balanceado.clase)'
```

```
ans = 1x2
      610      610
```

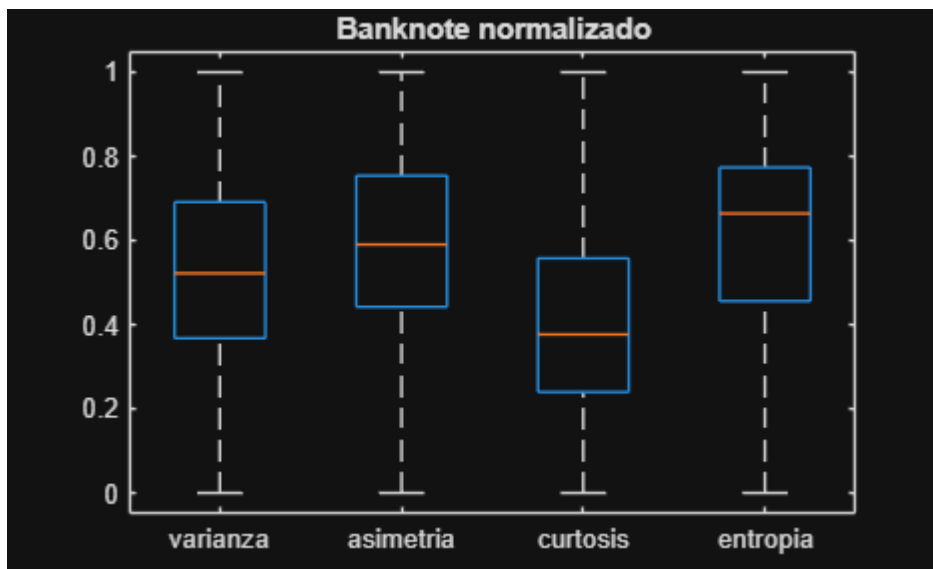
Paso 3: estandarizar

```
T3_estandarizado = T2_balanceado;
T3_estandarizado{:,vars} = zscore(T2_balanceado{:,vars});
figure
boxplot(T3_estandarizado{:,vars}, 'Labels', vars); title('Banknote estandarizado')
```



Paso 4: normalizar entre 0 y 1 (parte del balanceado)

```
T4_normalizado = T2_balanceado;
T4_normalizado[:,vars] = normalize(T2_balanceado[:,vars], 'range');
figure
boxplot(T4_normalizado[:,vars], 'Labels', vars); title('Banknote normalizado')
```



```
save('banknote_pasos.mat', 'T0_crudo', 'T1_outliers', 'T2_balanceado',
'T3_estandarizado', 'T4_normalizado')
```